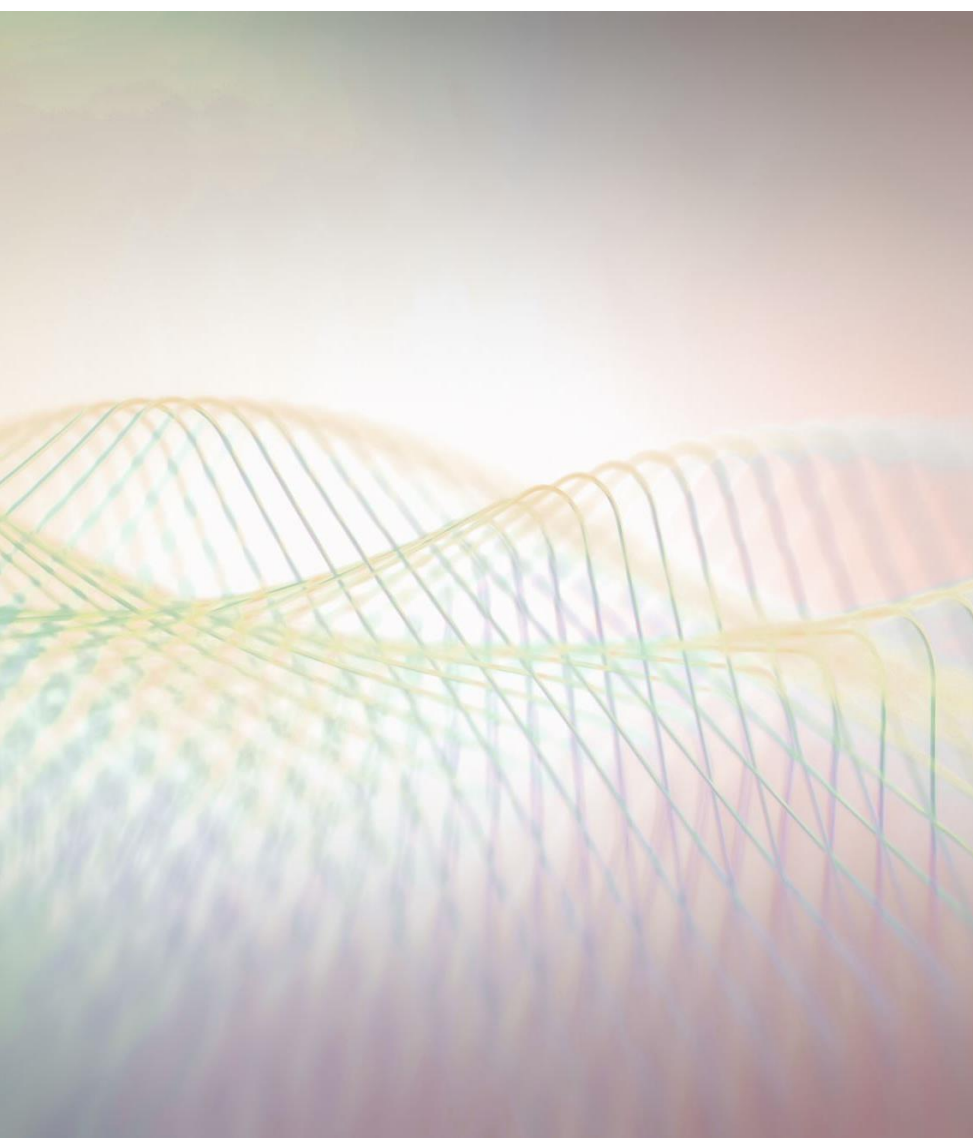




# COM2020

## AI Failures Museum: Learning from AI's Failures in Environmental Decision-Making

Java\_The\_Hutt Coursework 1 – Prototype v0.1.0

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# EXECUTIVE SUMMARY

- Educational Platform for users to learn from AI Failures
- Exhibits cover many fields
- Complete educational interactive quizzes
- Receive feedback from those quizzes

# PROBLEM FRAMING & CONTEXT

## 1. Context: AI Hype and Over-Reliance

- Rapid growth of AI adoption across industries
- Public and media focus often centres on breakthroughs rather than failures

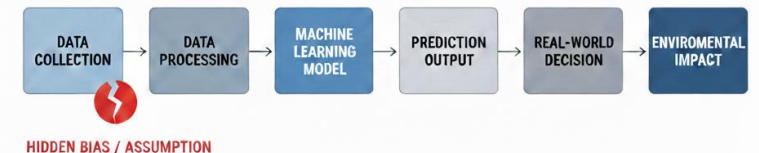
## 2. Fragmented Documentation of Failures

- High-profile AI failures are dispersed across news and research
- No structured platform consolidates these cases for educational use

## 3. Our Response

- Centralised repository of documented AI failure case studies
- Supporting artefacts, quizzes, and discussion features to promote critical engagement

### AI DECISION-MAKING PIPELINE & FAILURE POINTS



# PROTOTYPE WALKTHROUGH

— demonstrate —

# DATA DESIGN

- Data is created and modified in 2 JSON files:
- These are then seeded into a simple sqlite3 database, which is fetched from when the website loads
  - Exhibits.json
  - Quizzes.json
- > db.sqlite3

# SCOPE

**Primary goal:** Educate users about AI failures in environmental decision-making.

- Visitors can navigate, understand failures, complete quizzes, and see points/badges update.
- Curators can manage exhibits and supporting artefacts.
- Prototype works locally and online with seeded dataset

# TESTING

## Testing the Backend

- Authentication
- Web Server Health
- Account Configuration
- Creating, Modifying and Deleting Objects



[https://upload.wikimedia.org/wikipedia/commons/b/ba/Pytest\\_logo.svg](https://upload.wikimedia.org/wikipedia/commons/b/ba/Pytest_logo.svg)

# TESTING

```
===== test session starts =====
platform linux -- Python 3.11.14, pytest-9.0.2, pluggy-1.6.0
django: version: 5.2.10, settings: museum.settings (from ini)
rootdir: /workspaces/TeamProjectCOM2020/backend
configfile: pytest.ini
plugins: django-4.11.1
collected 28 items

exhibits/tests/test_account_creation.py .... [ 14%]
exhibits/tests/test_auth.py ..... [ 53%]
exhibits/tests/test_comments.py ..... [ 71%]
exhibits/tests/test_profile.py .. [ 78%]
exhibits/tests/test_web_server.py ..... [100%]

===== 28 passed in 10.66s =====
```



# TESTING

```
@pytest.mark.django_db
def test_sign_up_creates_user(client):
    response = client.post(reverse("signup"), {"username": "Alice",
                                                "password1": "Password!",
                                                "password2": "Password!"})

    assert response.status_code == 302
    assert User.objects.filter(username="Alice").exists()

@pytest.mark.django_db
def test_successful_login(client, user):

    response = client.post(reverse("login"), {"username": "Alice",
                                                "password": "Password1"})

    assert response.status_code == 302

@pytest.mark.django_db
def test_invalid_password(client, user):
    response = client.post(reverse("login"), {"username": "Alice",
                                                "password": "WrongPassword!"})

    assert response.status_code == 200
```

```
@pytest.mark.django_db
def test_health_endpoint(client):
    url = reverse("health")
    response = client.get(url)

    assert response.status_code == 200
    assert response.json() == {"status": "ok"}

@pytest.mark.django_db
def test_home_page_loads(client):
    response = client.get(reverse("home"))

    assert response.status_code == 200
    assert "AI Ethics & Failures Case Studies" in response.content.decode()

@pytest.mark.django_db
def test_performance(client):
    start = time.time()
    response = client.get(reverse("home"))
    duration = time.time() - start
    assert duration < 1
```

# ETHICS, LEGAL & LICENSING

- Addresses GDPR and Privacy
- No sensitive information
- Educational not competitive
- Misinformation Prevention
- Accessibility and inclusion
- Conclusion: Ethics Approval not required

# RISKS & MITIGATIONS

- Inappropriate user comments
- Comments sections need to be regulated
- Censorship should be implemented in sprint 2
- Exposure to AI failures
- Might promote users to try them out
- Case studies are for education and information shouldn't be used for malicious intent

# RISKS & MITIGATIONS

- Users can create their own login details
- Credentials need to be stored safely
- Ensuring the users privacy
- Information we upload must be accurate
- Prevent spreading false information
- Important to use legitimate sources when researching

# PROCESS EVIDENCE

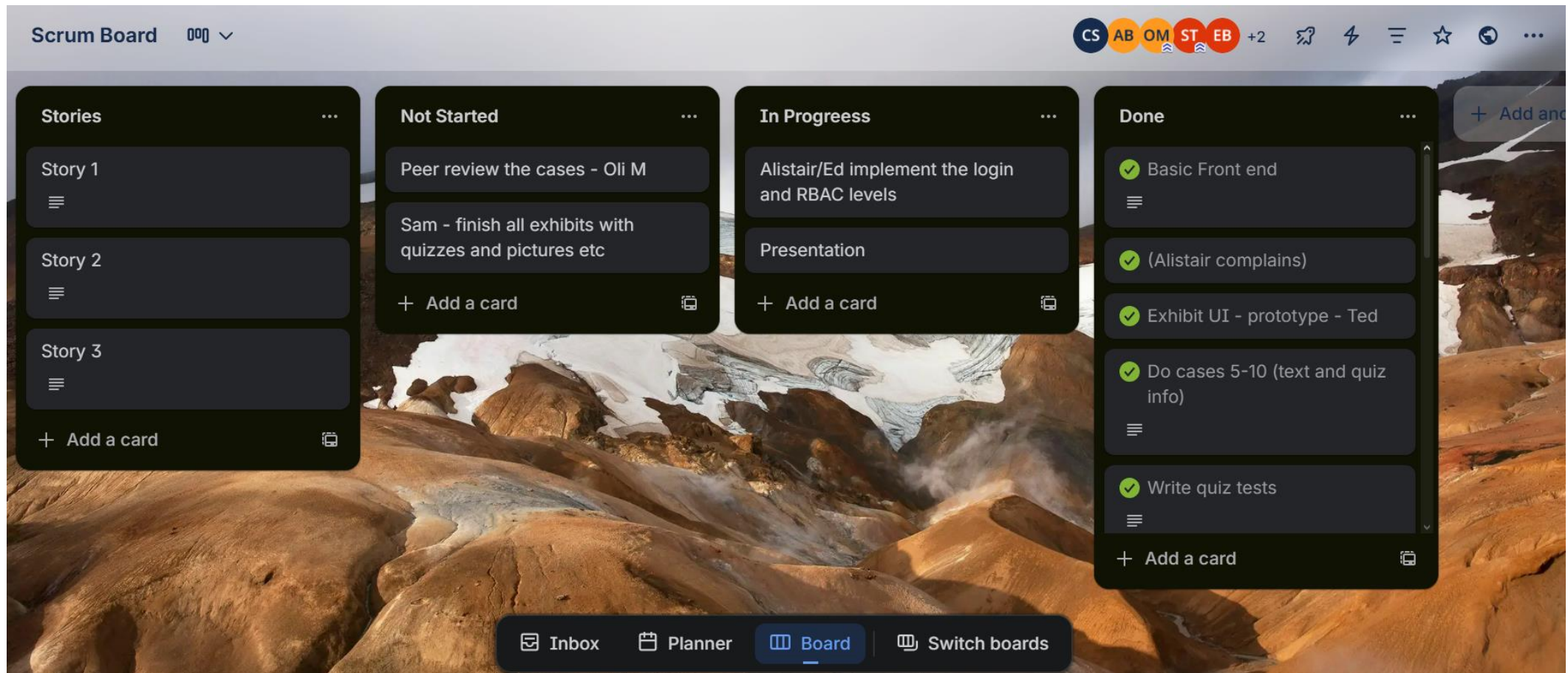
## Ways of Managing and Logging Meetings

- Scrum Board (Trello)
- Group Chat
- Emails
- Meeting Minutes and Summaries

## Weekly Discussions

- Reflection
- Future Steps
- Job Allocations
- Time Estimations
- Peer Help with Problems

# PROCESS EVIDENCE



# SPRINT 2

- Ability to add, delete and edit exhibits
- Deploy online
- Bookmarks
- Search Bar
- Improved UI
- Finalised Content